In class lab #2 for Numerical Analysis

Your name: _____

scan and submit

Part I Task 1 by MATLAB :

- Click on Editor and click on New to create a new document.
- Type the following piece of code.
- Once you are done, save it by the name `bisect.m`.

```
%Program 1.1 Bisection Method
%Compute approximate solution of f(x)=0
%Input: function handle f; a,b such that f(a)*f(b)<0,
%        and tolerance tol
%Output: Approximate solution xc
function xc=bisect(f,a,b,tol)
if sign(f(a))*sign(f(b)) >= 0
  error('f(a)f(b)< not satisfied!') %ceased execution
end
fa=f(a);
fb=f(b);
while (b-a)/2>tol
  c=(a+b)/2;
  fc=f(c);
  if fc == 0              %c is a solution, done
    break
  end
  if sign(fc)*sign(fa)<0  %a and c make the new interval
    b=c;fb=fc;
  else                    %c and b make the new interval
    a=c;fa=fc;
  end
end
xc = (a+b)/2;             %new midpoint is best estimate
end
```

The MATLAB you should know to understand this.

- What an m-file is and how the word `function` defines a function stored in a MATLAB file (an m-file)
- The meaning of the decission structures

```
if (conditions)
    ...
end
```

```
if (conditions)
    ...
else
    ...
end
```

```
if (conditions)
    ...
elseif (conditions)
    ...
else
    ...
end
```

Note that we write `elseif` and not `else if`. The latter means something different.

- How to use a `while` loop.
- How to use `break` to leave a `while` loop.
- The comparison symbol `==`.

Problem 1. (5 points) Type the following in the command window.

```
>> help bisect
```

What happens?

Problem 2. (20 points)Consider the equation $3x^3 + x^2 = x + 1$.

(a) Starting with $[0, 1]$, how many steps of the Bisection Method are required to calculate the solution with 0.5×10^{-10}? Answer with an integer. (Hint: after n steps of the Bisection method, solution error$= |x_c - r| < \frac{b_0 - a_0}{2^{n+1}}$.)

(b) Please use `bisect.m` to find a root of the function $3x^3 + x^2 = x + 1$ in the interval $[0, 1]$ to within 0.5×10^{-10}.

```
>> format long
>> f=@(x) 3*x^3+x^2-x-1;
>> xc=bisect( , , ,        )
%Please input function handle f and the other 3 parameters.

xc =
```

(c) Use the code `bisect.m` to find the root to **six** correct decimal places (with 0.5×10^{-8}) of the following equations

$$3x^3 + x^2 = x + 5 \qquad \text{and} \qquad \sin x = 6x + 5.$$

(You will **first need to figure out an interval** where you can guarantee the algorithm to find a solution.) Please **write down the code and their results**.

Fixed-Point Iteration

```
%Program 1.3 Fixed-Point Iteration
%Computes approximate solution of g(x)=x
%Input: function handle g, initial guess x0,
%        number of steps k
%Output: approximate solution xc
function xc=fpi(g,x0,k)
x(1)=x0;
for i = 1:k
  x(i+1) = g(x(i));
end
xc = x(k+1);
```

Problem 1. (5 points) Click on New to create a new document and type the above code. Once you are done, save it by the name `fpi.m`. Run the following code in the command window.

```
>> format long
>> g=@(x) (1+2*x^3)/(1+3*x^2);
>> xc=fpi(g,0.5,7)

xc =
```

```
%Program 1.3 Fixed-Point Iteration
%Computes approximate solution of g(x)=x
%Input: function handle g, initial guess x0,
%       the tolerance tol,
%       the maximum number of iterations maxIt
%Output: approximate solution xc
%        the number of iterations i
function [xc,i] = fpi2(g,x0,tol,maxIt)
x(1)=x0;
for i=1:maxIt-1
  x(i+1)=g(x(i));
  err=abs(x(i+1)-x(i));
  if (err<tol),break;end
end
xc=x(i+1);
i = i+1;
if i == maxIt
  disp('maximum number of iterations exceeded')
end
```

Problem 2. (10 points) Click on New to create a new document and type the above code. Once you are done, save it by the name `fpi2.m`.

Similar to Problem 1, write the code by using `fpi2.m` to iterate $g(x) = (1 + 2x^3)/(1 + 3x^2)$ by starting with an initial guess $x_0 = 0.5$. Tolerance should be set as 0.5×10^{-8}. maxIt can be set as 100. Output the approximate solution and number of steps needed.

Problem 3. (20 points) Define Fixed-Point Iteration on the equation $x^2 + x = \frac{5}{16}$ by isolating the x term. Determine the initial guesses by plotting $g(x)$ and drawing cobweb diagrams. Use `fpi2.m` to find both fixed points to **eight correct decimal** and state the number of steps needed.

Problem 4. (10 points) Please rewrite the equation $x^3 - x + e^x = 0$ into the form of $x = g(x)$ in four different ways.

Problem 5. (30 points) The equation $3x^3+2x^2+2x-1 = 0$ can be rewritten into the form of $x = g(x)$ in the following two different ways:

- $x = g(x), g(x) = \frac{-3x^3-2x^2+1}{2}$,
- $x = g(x), g(x) = \frac{-3x^3-2x^2+x+1}{3}$.

(a) Use Theorem 1.6 to determine whether Fixed-Point Iteration for each $g(x)$ is locally convergent to the given fixed point $r = \frac{1}{3}$.

(b) Run Fixed-Point Iteration for each g(x) with the initial point $x_0 = \frac{1}{2}$ and report the results of the first 10 iterations in tables. The tables should contain x_i, $g(x_i)$, $e_i = |x_i - r|$ and e_i/e_{i-1}, where $i = 1, 2, \cdots, 10$.